1 $|M|^2$ Process 1-7

1.1 Code

```
ClearAll["Global`*"];
Needs["FeynCalc`"];

denS[s_] := s - MA^2;
denT[t_] := t - MA^2;
denU[u_] := u - MA^2;

GetMandelstamRules[s_, t_, u_, m1_, m2_, m3_, m4_] := FCI[{
  SP[p1, p1] -> m1^2, SP[p2, p2] -> m2^2, SP[p3, p3] -> m3^2, SP[p4, p4] ->
  SP[p1, p2] -> (s - m1^2 - m2^2)/2, SP[p2, p1] -> (s - m1^2 - m2^2)/2,
  SP[p3, p4] -> (s - m3^2 - m4^2)/2, SP[p4, p3] -> (s - m3^2 - m4^2)/2,
  SP[p1, p3] -> (m1^2 + m3^2 - t)/2, SP[p3, p1] -> (m1^2 + m3^2 - t)/2,
  SP[p2, p4] -> (m2^2 + m4^2 - t)/2, SP[p4, p2] -> (m2^2 + m4^2 - t)/2,
  SP[p1, p4] -> (m1^2 + m4^2 - u)/2, SP[p4, p1] -> (m1^2 + m4^2 - u)/2,
  SP[p2, p3] -> (m2^2 + m3^2 - u)/2, SP[p3, p2] -> (m2^2 + m3^2 - u)/2
}];

ProcessCleanUp[rawMatrix_, rule_, elimRule_] :=
  rawMatrix // ChangeDimension[#, 4] & // ExpandScalarProduct //
  ReplaceAll[#, rule] & // ReplaceAll[#, elimRule] & // Simplify;

(*---Process 1: \[Chi] \[Chi]bar -> \[Chi] \[Chi]bar (s,t)---*)
rule1 = GetMandelstamRules[s, t, u, mchi, mchi, mchi, mchi];
M1s = gchi^2/denS[s] * SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]
M1t = gchi^2/denT[t] * SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]
M1 = M1s - M1t;
M1sqRaw = FermionSpinSum[M1 * ComplexConjugate[M1]]/4 // Contract // DiracS
M1sq = ProcessCleanUp[M1sqRaw, rule1, u -> 4 mchi^2 - s - t];

(*---Process 2: \[Chi] \[Chi] -> \[Chi] \[Chi] (t,u)---*)
M2t = gchi^2/denT[t] * SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]
M2u = gchi^2/denU[u] * SpinorUBar[p4, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]
M2 = M2t - M2u;
M2sqRaw = FermionSpinSum[M2 * ComplexConjugate[M2]]/4 // Contract // DiracS
M2sq = ProcessCleanUp[M2sqRaw, rule1, s -> 4 mchi^2 - t - u];

(*---Process 3: \[Chi]bar \[Chi]bar -> \[Chi]bar \[Chi]bar (t,u)---*)
M3t = gchi^2/denT[t] * SpinorVBar[p1, mchi] . GA[\[Mu]] . SpinorV[p3, mchi]
M3u = gchi^2/denU[u] * SpinorVBar[p1, mchi] . GA[\[Mu]] . SpinorV[p4, mchi]
M3 = M3t - M3u;
M3sqRaw = FermionSpinSum[M3 * ComplexConjugate[M3]]/4 // Contract // DiracS
M3sq = ProcessCleanUp[M3sqRaw, rule1, s -> 4 mchi^2 - t - u];

(*---Process 4: \[Mu]- \[Mu]+ -> \[Chi] \[Chi]bar---*)
```

```
rule4 = GetMandelstamRules[s, t, u, mmu, mmu, mchi, mchi];
M4 = (gmu * gchi) / denS[s] * SpinorVBar[p2, mmu] . GA[\[Mu]] . SpinorU[p1,
M4sqRaw = FermionSpinSum[M4 * ComplexConjugate[M4]]/4 // Contract // DiracS
M4sq = ProcessCleanUp[M4sqRaw, rule4, {}];

(*---Process 5: \[Tau]- \[Tau]+ -> \[Chi] \[Chi]bar---*)
rule5 = GetMandelstamRules[s, t, u, mtau, mtau, mchi, mchi];
M5 = (gtau * gchi) / denS[s] * SpinorVBar[p2, mtau] . GA[\[Mu]] . SpinorU[p
M5sqRaw = FermionSpinSum[M5 * ComplexConjugate[M5]]/4 // Contract // DiracS
M5sq = ProcessCleanUp[M5sqRaw, rule5, {}];

(*---Process 6: \[Nu]_\[Mu] vbar_\[Mu] -> \[Chi] \[Chi]bar---*)
rule6 = GetMandelstamRules[s, t, u, 0, 0, mchi, mchi];
M6 = (gnumu * gchi) / denS[s] * SpinorVBar[p2, 0] . GA[\[Mu]] . ((1 - GA[5]
  SpinorU[p1, 0] * SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorV[p4, mchi];
M6sqRaw = FermionSpinSum[M6 * ComplexConjugate[M6]]/1 // Contract // DiracS
M6sq = ProcessCleanUp[M6sqRaw, rule6, {}];

(*---Process 7: \[Nu]_\[Tau] vbar_\[Tau] -> \[Chi] \[Chi]bar---*)
rule7 = GetMandelstamRules[s, t, u, 0, 0, mchi, mchi];
M7 = (gnutau * gchi) / denS[s] * SpinorVBar[p2, 0] . GA[\[Mu]] . ((1 - GA[5
  SpinorU[p1, 0] * SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorV[p4, mchi];
M7sqRaw = FermionSpinSum[M7 * ComplexConjugate[M7]]/1 // Contract // DiracS
M7sq = ProcessCleanUp[M7sqRaw, rule7, {}];

Print["Process1 (\[Chi] \[Chi]bar -> \[Chi] \[Chi]bar) : ", M1sq // Apart[#
Print["Process2 (\[Chi] \[Chi] -> \[Chi] \[Chi]) : ", M2sq // Apart[#, {t,
Print["Process3 (\[Chi]bar \[Chi]bar -> \[Chi]bar \[Chi]bar) : ", M3sq // A
Print["Process4 (\[Mu]- \[Mu]+ -> \[Chi] \[Chi]bar) : ", M4sq];
Print["Process5 (\[Tau]- \[Tau]+ -> \[Chi] \[Chi]bar) : ", M5sq];
Print["Process6 (v_\[Mu] v_\[Mu]bar -> \[Chi] \[Chi]bar) : ", M6sq];
Print["Process7 (v_\[Tau] v_\[Tau]bar -> \[Chi] \[Chi]bar) : ", M7sq];
```

1.2 Results

Process 1 ($\chi\bar{\chi} \to \chi\bar{\chi}$) :

$$\left\{ \frac{4g_\chi^4 s^2}{(M_A^2-t)^2} - \frac{4g_\chi^4 s(M_A^2+4m_\chi^2-2t)}{(M_A^2-t)^2} - \frac{2g_\chi^4(M_A^4-8m_\chi^4+4M_A^2t-6t^2)}{(M_A^2-t)^2} \right.$$
$$- \frac{8(-2g_\chi^4 m_\chi^4+g_\chi^4 M_A^2 t+g_\chi^4 t^2)}{(-M_A^2+s)(M_A^2-t)} + \frac{2(g_\chi^4 M_A^4+8g_\chi^4 m_\chi^4+2g_\chi^4 M_A^2 t-8g_\chi^4 m_\chi^2 t+2g_\chi^4 t^2)}{(-M_A^2+s)^2},$$
$$- \frac{2g_\chi^4(M_A^4-8m_\chi^4+4M_A^2 s-6s^2)}{(M_A^2-s)^2} - \frac{4g_\chi^4(M_A^2+4m_\chi^2-2s)t}{(M_A^2-s)^2} + \frac{4g_\chi^4 t^2}{(M_A^2-s)^2}$$
$$\left. + \frac{2(g_\chi^4 M_A^4+8g_\chi^4 m_\chi^4+2g_\chi^4 M_A^4 m_\chi^2 s+2g_\chi^4 s^2)}{(-M_A^2+t)^2} - \frac{8(-2g_\chi^4 m_\chi^4+g_\chi^4 M_A^2 s+g_\chi^4 s^2)}{(M_A^2-s)(-M_A^2+t)} \right\}$$

Process 2 $(\chi\chi \to \chi\chi)$:

$$\left\{ \frac{4g_\chi^4 t^2}{(M_A^2-u)^2} - \frac{4g_\chi^4 t(M_A^2+4m_\chi^2-2u)}{(M_A^2-u)^2} - \frac{2g_\chi^4(M_A^4-8m_\chi^4+4M_A^2u-6u^2)}{(M_A^2-u)^2} \right.$$
$$- \frac{8(-2g_\chi^4 m_\chi^4+g_\chi^4 M_A^2 u+g_\chi^4 u^2)}{(-M_A^2+t)(M_A^2-u)} + \frac{2(g_\chi^4 M_A^4+8g_\chi^4 m_\chi^4+2g_\chi^4 M_A^2 u-8g_\chi^4 m_\chi^2 u+2g_\chi^4 u^2)}{(-M_A^2+t)^2},$$
$$+ \frac{2g_\chi^4(M_A^4-8m_\chi^4+4M_A^2 t-6t^2)}{(M_A^2-t)^2} - \frac{4g_\chi^4(M_A^2+4m_\chi^2-2t)u}{(M_A^2-t)^2} + \frac{4g_\chi^4 u^2}{(M_A^2-t)^2}$$
$$\left. + \frac{2(g_\chi^4 M_A^4+8g_\chi^4 m_\chi^4+2g_\chi^4 M_A^2 t-8g_\chi^4 m_\chi^2 t+2g_\chi^4 t^2)}{(-M_A^2+u)^2} - \frac{8(-2g_\chi^4 m_\chi^4+g_\chi^4 M_A^2 t+g_\chi^4 t^2)}{(M_A^2-t)(-M_A^2+u)} \right\}$$

Process 3 $(\bar{\chi}\bar{\chi} \to \bar{\chi}\bar{\chi})$:

$$\left\{ \frac{4g_\chi^4 t^2}{(M_A^2-u)^2} - \frac{4g_\chi^4 t(M_A^2+4m_\chi^2-2u)}{(M_A^2-u)^2} - \frac{2g_\chi^4(M_A^4-8m_\chi^4+4M_A^2u-6u^2)}{(M_A^2-u)^2} \right.$$
$$- \frac{8(-2g_\chi^4 m_\chi^4 u+g_\chi^4 u^2)}{(-M_A^2+t)(M_A^2-u)} + \frac{2(g_\chi^4 M_A^4+8g_\chi^4 m_\chi^4+2g_\chi^4 M_A^2 u-8g_\chi^4 m_\chi^2 u+2g_\chi^4 u^2)}{(-M_A^2+t)^2},$$
$$+ \frac{2g_\chi^4(M_A^4-8m_\chi^4+4M_A^2 t-6t^2)}{(M_A^2-t)^2} - \frac{4g_\chi^4(M_A^2+4m_\chi^2-2t)u}{(M_A^2-t)^2} + \frac{4g_\chi^4 u^2}{(M_A^2-t)^2}$$
$$\left. + \frac{2(g_\chi^4 M_A^4+8g_\chi^4 m_\chi^4+2g_\chi^4 M_A^2 t-8g_\chi^4 m_\chi^2 t+2g_\chi^4 t^2)}{(-M_A^2+u)^2} - \frac{8(-2g_\chi^4 m_\chi^4+g_\chi^4 M_A^2 t+g_\chi^4 t^2)}{(M_A^2-t)(-M_A^2+u)} \right\}$$

Process 4 $(\mu^-\mu^+ \to \chi\bar{\chi})$: $\dfrac{2g_\chi^2 g_\mu^2(2m_\chi^4+2m_\mu^4+t^2+2m_\mu^2(s-t-u)+2m_\chi^2(2m_\mu^2+s-t-u)+u^2)}{(M_A^2-s)^2}$

Process 5 $(\tau^-\tau^+ \to \chi\bar{\chi})$: $\dfrac{2g_\chi^2 g_\tau^2(2m_\chi^4+2m_\tau^4+t^2+2m_\tau^2(s-t-u)+2m_\chi^2(2m_\tau^2+s-t-u)+u^2)}{(M_A^2-s)^2}$

Process 6 $(\nu_\mu\bar{\nu}_\mu \to \chi\bar{\chi})$: $\dfrac{4g_\chi^2 g_{\nu_\mu}^2(2m_\chi^4+t^2+2m_\chi^2(s-t-u)+u^2)}{(M_A^2-s)^2}$

Process 7 $(\nu_\tau\bar{\nu}_\tau \to \chi\bar{\chi})$: $\dfrac{4g_\chi^2 g_{\nu_\tau}^2(2m_\chi^4+t^2+2m_\chi^2(s-t-u)+u^2)}{(M_A^2-s)^2}$

1.3 Checks

1.3.1 Process 1

$$|M|^2 = \frac{4g_\chi^4 s^2}{(M_A^2-t)^2} - \frac{4g_\chi^4 s(M_A^2+4m_\chi^2-2t)}{(M_A^2-t)^2} - \frac{2g_\chi^4(M_A^4-8m_\chi^4+M_A^2t-6t^2)}{(M_A^2-t)^2} - \frac{8(-2g_\chi^4 m_\chi^4+g_\chi^4 M_A^2 t+g_\chi^4 t^2)}{(-m_\chi^2+s)(M_A^2-t)} + \frac{2(2g_\chi^4 m_\chi^4+2g_\chi^4 u^2-2g_\chi^4 u^2+\dots)}{(-m_\chi^2+s)^2}$$

Let: $\chi \to e$, $m_\chi \to m_e \to 0$, $g_\chi \to g_e = e$, $M_A \to M_\gamma = 0$:

$$\begin{aligned}
|M|^2 &= \frac{4e^4 s^2}{t^2} - \frac{4e^4 s(-2t)}{t^2} - \frac{2e^4(-6t^2)}{t^2} - \frac{8(e^4t^2)}{-st} + \frac{2(2e^4t^2)}{s^2} \\
&= \frac{4e^4s^2+8e^4st+12e^4t^2}{t^2} + \frac{8e^4t^2}{st} + \frac{4e^4t^2}{s^2} \\
&= \frac{4e^4(s^4+2s^3t+3s^2t^2+2st^3+t^4)}{s^2t^2} \\
&= \frac{2e^4(s^4+t^4+(s+t)^4)}{s^2t^2}
\end{aligned}$$

Since $s+t+u = \sum m_i^2 = 0 \implies u = -(s+t)$, we have $(s+t)^4 = u^4$:

$$\begin{aligned}
|M|^2 &= \frac{2e^4(s^4+t^4+u^4)}{s^2t^2} \\
&= \frac{2e^4(s^4+t^4+u^2(s+t)^2)}{s^2t^2} \\
&= \frac{2e^4(s^4+t^4+u^2s^2+u^2t^2+2u^2st)}{s^2t^2}
\end{aligned}$$

Same as what it shows on FeynCalc website for Bhabha scattering.

1.3.2 Process 2

Let: $\chi \to e$, $m_\chi \to m_e \to 0$, $M_A \to M_\gamma = 0$, $g_\chi \to g_e = e$:

$$\begin{aligned}
|M|^2 &= \frac{4e^4t^2}{u^2} - \frac{4e^4t(-2u)}{u^2} - \frac{2e^4(-6u^2)}{u^2} - \frac{8(e^4u^2)}{-tu} + \frac{2(2e^4u^2)}{t^2} \\
&= 2e^4\frac{2t^2+4ut+6u^2}{u^2} + \frac{8e^4u^2}{tu} + \frac{4e^4u^2}{t^2} \\
&= \frac{2e^4(2t^4+4ut^3+6u^2t^2+4u^3t+2u^4)}{u^2t^2} \\
&= \frac{2e^4(t^4+u^4+(t+u)^4)}{u^2t^2} \\
&= \frac{2e^4(s^2(t+u)^2+t^4+u^4)}{t^2u^2}
\end{aligned}$$

Same as FeynCalc website for Møller scattering $e^-e^- \to e^-e^-$.

1.3.3 Process 3

Same as Process 2.

1.3.4 Process 5 ($\tau^-\tau^+ \to \chi\bar{\chi}$)

Let: $\chi \to \mu$, $g_\chi \to g_\mu = e$, $\tau \to e$, $A' \to \gamma$, $g_\tau \to g_e = e$, $m_\tau \to m_e \to 0$, $m_\chi \to m_\mu$, $M_A \to M_\gamma = 0$:

$$\begin{aligned} |M|^2 &= \frac{2e^2e^2(2m_\mu^4 + t^2 + 2m_\mu^2(s-t-u) + u^2)}{s^2} \\ &= \frac{2e^4(2m_\mu^4 + 2m_\mu^2(s-t-u) + t^2 + u^2)}{s^2} \end{aligned}$$

Same as FeynCalc website.

2 Transfer Cross Section σ_T

Use $\sigma_T = \int d\Omega(1-\cos\theta)\frac{d\sigma}{d\Omega}$ for Process 1, 2, 3.

$$p_{cm}^2 = \frac{s-4m_\chi^2}{4} \tag{1}$$

$$z = cos\theta \tag{2}$$

$$t = -2p_{cm}^2(1-cos\theta) = -\frac{s-4m_\chi^2}{2}(1-z) \tag{3}$$

$$u = -2p_{cm}^2(1+cos\theta) = -\frac{s-4m_\chi^2}{2}(1+z) \tag{4}$$

$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s}|M|^2$

$\sigma_T = -\int_1^{-1} dz \frac{d\sigma}{d\Omega}(1-z)2\pi = 2\pi\int_{-1}^{1} dz \frac{d\sigma}{d\Omega}(1-z)$

For Process 1, output distinguishable, but for Process 2&3 output identical. So there should be an 1/2 correction. Because when integrating through θ, will double count output particles at $\theta\&\pi-\theta$ and $\pi-\theta\&\theta$.

2.1 Code

```
(*   z = cos\[Theta] *)
kinematicRules = {
  t -> -(s - 4 mchi^2)/2 * (1 - z),
  u -> -(s - 4 mchi^2)/2 * (1 + z)
};


dSigmaDz[Msq_] := (Msq /. kinematicRules) / (32 Pi s);


(* Process 1: *)
SigmaT1 = Integrate[dSigmaDz[M1sq] * (1 - z), {z, -1, 1},
   Assumptions -> {s > 4 mchi^2, MA > 0, mchi > 0}] // Simplify;
Print["\[Sigma]_T (Process 1) = ", SigmaT1];
```

```
(* Process 2: *)
SigmaT2 = 1/2 * Integrate[dSigmaDz[M2sq] * (1 - z), {z, -1, 1},
    Assumptions -> {s > 4 mchi^2, MA > 0, mchi > 0}] // Simplify;
Print["\[Sigma]_T (Process 2) = ", SigmaT2];

(* Process 3: *)
SigmaT3 = 1/2 * Integrate[dSigmaDz[M3sq] * (1 - z), {z, -1, 1},
    Assumptions -> {s > 4 mchi^2, MA > 0, mchi > 0}] // Simplify;
Print["\[Sigma]_T (Process 3) = ", SigmaT3];
```

2.2 Results

$$\sigma_T^{(\text{Process 1})} = -\frac{g_\chi^4}{24\pi(M_A^2-s)^2 s(s-4m_\chi^2)^2}\times$$
$$\left[6(M_A^2-s)\Big(M_A^6-3sM_A^4+4(4m_\chi^4-2sm_\chi^2-s^2)M_A^2-2s(s-2m_\chi^2)^2\Big)\ln\left(\frac{M_A^2}{M_A^2-4m_\chi^2+s}\right)\right.$$
$$-\frac{4m_\chi^2-s}{M_A^2-4m_\chi^2+s}\Big(6M_A^8-3(4m_\chi^2+7s)M_A^6+(80m_\chi^4+8sm_\chi^2-19s^2)M_A^4$$
$$\left.+(96m_\chi^6-8sm_\chi^4+76s^2m_\chi^2+7s^3)M_A^2-128m_\chi^8+23s^4-64m_\chi^2s^3-24m_\chi^4s^2-128m_\chi^6 s\Big)\right]$$

$$\sigma_T^{(\text{Process 2})} = -\frac{g_\chi^4}{16\pi s}\left[\frac{2}{M_A^2}\left(-\frac{M_A^4+8m_\chi^4}{M_A^2-4m_\chi^2+s}-2s\right)\right.$$
$$\left.+\frac{4\Big(2M_A^4+(3s-4m_\chi^2)M_A^2+4m_\chi^2(s-3m_\chi^2)\Big)}{(4m_\chi^2-s)(2M_A^2-4m_\chi^2+s)}\ln\left(\frac{M_A^2}{M_A^2-4m_\chi^2+s}\right)-2\right]$$

The result for Process 3 ($\bar{\chi}\bar{\chi}\to\bar{\chi}\bar{\chi}$) is identical to Process 2:

$$\sigma_T^{(\text{Process 3})} = \sigma_T^{(\text{Process 2})}$$

2.3 Checks

I cannot find literatures. So I tried to take limits first and see it the results collapse to something.

3 NonRelativistic Limit

In CM frame, $s=(E_1+E_2)^2$, $p_1+p_2=0$,$E=\sqrt{m_\chi^2+|p|^2}$

So, $s=(2E)^2=4(m_\chi^2+|p|^2)=4m_\chi^2+4|p|^2=4m_\chi^2+4m_\chi^2v_{CM}^2$

$v=v_{rel}=2v_{CM}$, so $s=4m_\chi^2+m_\chi^2v^2$

$\alpha_\chi=\frac{g_\chi^2}{4\pi}$

3.1 code

```
NRRule = {s -> 4 mchi^2 + mchi^2 v^2};
AlphaRule = {gchi^4 -> 16 Pi^2 alphaX^2};

SigmaT1NR = SigmaT1 /. NRRule /. AlphaRule;
SigmaT1Leading = Series[SigmaT1NR, {v, 0, 0}] // Normal // FullSimplify;
Print[" _T  (Process 1) = "];
Print[SigmaT1Leading];

SigmaT2NR = SigmaT2 /. NRRule /. AlphaRule;
SigmaT2Leading = Series[SigmaT2NR, {v, 0, 0}] // Normal // FullSimplify;
Print[" _T  (Process 2) = "];
Print[SigmaT2Leading];
```

3.2 Results

$s \simeq 4m_\chi^2 + m_\chi^2 v^2$

$\alpha_X = \frac{g_\chi^2}{4\pi}$,

$v \to 0$, 0th order.

$$\sigma_T^{(\text{Process 1})}\Big|_{v\to 0} \cong \frac{4\pi\alpha_X^2 m_\chi^2(7M_A^4 - 20M_A^2 m_\chi^2 + 16m_\chi^4)}{M_A^4(M_A^2 - 4m_\chi^2)^2}$$

$$\sigma_T^{(\text{Process 2})}\Big|_{v\to 0} \cong \frac{2\pi\alpha_X^2 m_\chi^2}{M_A^4}$$

3.3 Analysis

If $M_A^2 \cong 4m_\chi^2, M_A \cong 2m_\chi$
$\sigma_T^{(\text{Process 1})}$ diverges. So Transfer Cross Section in NR Limit and specific mass ratio will be very significant.

If $M_A << m_\chi, \frac{M_A}{m_\chi} = \varepsilon$,

$$\sigma_T^{(\text{Process 1})}\Big|_{v\to 0} \cong \frac{4\pi\alpha_X^2 m_\chi^2(7M_A^4 - 20M_A^2 m_\chi^2 + 16m_\chi^4)}{M_A^4(M_A^2 - 4m_\chi^2)^2} \tag{5}$$

$$= \frac{4\pi\alpha_\chi^2 m_\chi^2(7\varepsilon^4 - 20\varepsilon^2 + 16)}{M_A^4(\varepsilon^2 - 4)^2} \tag{6}$$

$$\cong \frac{4\pi\alpha_\chi^2 m_\chi^2 \cdot 16}{M_A^4 \cdot 16} \tag{7}$$

$$= \frac{4\pi\alpha_\chi^2 m_\chi^2}{M_A^4} \simeq 2\sigma_T^{(\text{Process 2})} \tag{8}$$

In Appendix B of 1709.07882 [1], $\sigma_T^{\text{Born}} \cong \frac{8\pi\alpha_\chi^2}{m_\chi^2 v^4}(\ln(1+R^2)+\frac{R^2}{1+R^2}), R = \frac{m_\chi v}{m_\varphi}$. φ is the mediator, here is A'
When $R << 1, v \to 0$,

$$\ln(1+R^2) - \frac{R^2}{1+R^2} \tag{9}$$
$$\cong R^2 - \frac{R^4}{2} + \frac{R^6}{3} + \cdots - \frac{1}{\frac{1}{R^2}+1} \tag{10}$$
$$\cong R^2 - \frac{R^4}{2} - (1+\frac{1}{R^2})^{-1} \tag{11}$$
$$\cong R^2 - \frac{R^4}{2} - [R^2 - (R^2)^2 + (R^2)^3 + \ldots] \tag{12}$$
$$\cong R^2 - \frac{R^4}{2} - R^2 + R^4 + \ldots \tag{13}$$
$$\cong \frac{R^4}{2} \tag{14}$$
$$\tag{15}$$

$$\sigma_T^{\text{Born}} \cong \frac{8\pi\alpha_\chi^2}{m_\chi^2 v^4} \cdot \frac{R^4}{2} \tag{16}$$
$$= \frac{4\pi\alpha_\chi^2}{m_\chi^2 v^4} \cdot \frac{m_\chi^4 v^4}{M_A^4} \tag{17}$$
$$= \frac{4\pi\alpha_\chi^2 m_\chi^2}{M_A^4} = \sigma_T^{(\text{Process 1})}(When\ M_A << m_\chi) \tag{18}$$

In this case ($M_A << m_\chi$), $\frac{m_\chi}{M_A} >> 1, v << 1$, should discuss if R still $<< 1$? If still make sense, my result corresponds to literature's.

For Processes 2 & 3, it is one half the result, because the phase space integral I had an $\frac{1}{2}$ factor.

The paper 1210.0900's appendix [2] claims "in the Born regime, for both attractive and repulsive forces, a perturbative calculation gives $\sigma_T^{\text{Born}} \cong \frac{8\pi\alpha_\chi^2}{m_\chi^2 v^4}(\ln(1+R^2)+\frac{R^2}{1+R^2})$

Processes 2 & 3 are repulsive interactions. Should be half of attractive interactions.

So for the coefficient 8, I guess the correct σ_T^{Born} should be $\frac{4\pi\alpha_\chi^2}{m_\chi^2 v^4}(\ln(1+R^2)+\frac{R^2}{1+R^2})$.
The reason is not there is a different potential, the core is identical particles. So just divide by 2! for the potential's perturbation results.

[1] Simon Knapen, Tongyan Lin and Kathryn M. Zurek, arXiv 1709.07882
[2] Sean Tulin, Hai-Bo Yu, and Kathryn M. Zurek, arXiv:1210.0900

4 Comments

1, I didn't use the viscosity cross section $\int \frac{d\sigma}{d\Omega} d\Omega \sin^2\theta$ mentioned in paper 1709.07882 [3]. For identical particles, I used $\frac{1}{2}\sigma_T$. If use viscosity cross section, answers may be different.

2, In the neutrino average part, I divided by 1 instead of 4. If want to divide by 4, because it is linear process, I can just divide by 4 in the end. But it doesn't influence transfer cross section for processes 1-3.

3, Do I need to reference in this kind of research notes?

[3] Simon Knapen, Tongyan Lin and Kathryn M. Zurek, arXiv 1709.07882